

Module 2 — Quiz B (Concept Check)

Question bank · After Read 2 · open notes · unlimited attempts

Module 2, Quiz B: Concept check

Reading: [Module 2 Read 2](#) (complete Read 1 + Quiz A first)

Canvas: Concept Check quiz (Module 2)

Apply sampling designs, FPC decisions, z-scores, variable types, and mean/median ideas from Read 1–2.

Full CI word problems, R output, and error scenarios → [Synthesis](#).

Section A — Convenience vs SRS & bias (Read 1 §4–5)

A1. Dining hall intercept

First 60 students outside one dining hall 12–1 p.m. surveyed about sleep.

1. Convenience or SRS? 2. Bias likely? 3. Generalizability?

A2. Registrar SRS

Random-number draw of 120 unique IDs from full undergraduate registrar list.

1. Convenience or SRS? 2. Bias likely? 3. Generalizability?

A3. Club meeting

40 students at pre-med club meeting surveyed about med-school plans.

1. Convenience or SRS? 2. Bias likely? 3. Generalizability?

A4. Incomplete frame

SRS from declared-majors list only; undeclared students excluded.

1. Design label? 2. Bias? 3. Generalizability?

Answer key — Section A — Convenience vs SRS & bias (Read 1 §4–5)

1. **Convenience. Yes** — time/location sample differs from all students. **Limited** generalizability.
 2. **SRS. Low bias** if list matches population and response is good. **Reasonable** to generalize to list population.
 3. **Convenience. Yes** — club members differ from all undergraduates. **Poor** campus-wide generalizability.
 4. **SRS from incomplete frame. Yes** — undercoverage of undeclared. Generalize only to students **on the frame**.
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Section B — SRS steps & interval type

B1. SRS workflow

List three ordered steps for **simple random sampling**, then apply them to estimate mean study hours for $N = 210$ STAT 109 students with $n = 30$.

B2. Large-state proportion

$n = 220$, $x = 82$ with part-time job; SRS from very large statewide population.

Mean or proportion? FPC? Conditions?

B3. Finite classroom mean

$N = 300$ classrooms, SRS $n = 30$ without replacement; mean chairs.

Mean or proportion? FPC? Conditions?

B4. Clinic proportion with FPC

$N = 80$ patients, $n = 50$ without replacement; Yes/No recovery proportion.

Mean or proportion? FPC? Conditions?

B5. Small count proportion

Large campus SRS; biology major indicator; $\hat{p} = 2/18$.

Mean or proportion? FPC? Conditions?

Answer key — Section B — SRS steps & interval type

- 1) Number frame 1–210. 2) Draw 30 unique random IDs. 3) Collect data from those students.
 2. **Proportion. No FPC.** Likely OK: SRS + large counts ($n\hat{p}$ and $n(1 - \hat{p})$ both ≥ 10).
 3. **Mean. Use FPC** ($n/N = 0.10$). Random sample; check normality or moderate n .
 4. **Proportion. Use FPC** (large sampling fraction). Check large-count condition with observed x .
 5. **Proportion. No FPC** (large campus). **Large-count fails** ($x = 2 < 10$) — Wald normal CI questionable; consider exact/binomial methods.
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Section C — z-scores & statistics toolbox (Read 2 §1–2)

C1. Annarose z-score

Women: $\mu = 64$, $\sigma = 2.5$ in. Annarose is 67 in. Compute her z-score.

C2. Compare z-scores

Newton: 72 in; men $\mu = 70$, $\sigma = 3$. Who is more unusually tall relative to their group — Annarose ($z = 1.2$) or Newton?

C3. Standard normal probability

For $Z \sim N(0, 1)$, what does $P(Z \leq 1.24)$ represent? Also find $P(Z > 1.24)$ (table or calculator).

C4. Coin test hypotheses

16 heads in 20 flips; test if coin is fair (two-sided). State H_0 , H_a , and define p in words.

C5. Role of p_0

In $z = (\hat{p} - p_0) / \sqrt{p_0(1 - p_0)/n}$, what is p_0 ?

C6. `prop.test()` decision

When using `prop.test()` for a hypothesis test, which output value do you compare to α ?

C7. Descriptive vs inferential

One binary variable from a sample: give one **descriptive** summary and one **inferential** question answered by a CI.

C8. Variable types

Classify each as numerical or categorical (and binary if categorical):

- a) Stem diameter (cm)
- b) Soil type: sand/loam/clay
- c) Parasite present: yes/no
- d) Egg count per nest

Answer key — Section C — z-scores & statistics toolbox (Read 2 §1–2)

1. $z = (67 - 64)/2.5 = 1.2$ — 1.2 SD above the women's mean.
 2. Newton: $z = (72 - 70)/3 \approx 0.67$. **Annarose** (larger z-score).
 3. Proportion **at or below** 1.24 (about 0.893). $P(Z > 1.24) \approx 0.107$.
 4. $H_0 : p = 0.5$, $H_a : p \neq 0.5$. p = long-run probability of heads on one flip.
 5. **(b)** The population proportion **assumed under** H_0 for the standard error.
 6. The **p-value** — reject H_0 if p-value $< \alpha$.
 7. Descriptive: $\hat{p}$ or counts. Inferential: plausible **range for population proportion**.
 8. a) Numerical. b) Categorical, not binary. c) Categorical, binary. d) Numerical.
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Section D — Mean and median by hand (Read 2 §3)

D1. Mean and median

Data: 3, 5, 5, 7, 9. Compute the **mean** and **median**.

Answer key — Section D — Mean and median by hand (Read 2 §3)

1. Median = 5. Mean = $(3 + 5 + 5 + 7 + 9)/5 = 5.8$.
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